How Much of a Dark Matter Signal Is Really Cosmic Rays?

Muon-Induced Background and Veto Efficiency in the DM-Ice17 Signal Region

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1. Objective, Significance, and Implications

For more than two decades the DAMA/LIBRA experiment in Italy has reported that the low-energy event rate in its sodium-iodide dark matter detector rises and falls once a year, peaking near June 2, and has interpreted this as the Earth's orbit sweeping through a galactic halo of dark matter particles. No other experiment has reproduced it, and the central objection is mundane: laboratories also have seasons. Temperature, radon, and cosmic-ray muon flux all vary annually, and any of them could imprint a yearly cycle on such a detector.

DM-Ice17 is the instrument built to attack that ambiguity. It is a pair of 8.5 kg thallium-doped NaI crystals frozen into the Antarctic ice at 2450 m depth at the bottom of two IceCube Neutrino Observatory strings, and it is the only NaI dark matter detector ever operated inside a cubic-kilometre muon telescope—which means its dominant seasonal background can be tagged independently, event by event, by the surrounding array.

Over the past summer I established that the muon flux through DM-Ice modulates by $10.2\% \pm 1.6\%$, peaking in late January, 13.4σ away from the DAMA phase, and I recovered a muon selection for the 2015–2020 high-gain era that no published analysis had reached. But the muon flux is not what a dark matter search sees. Muons deposit far too much energy to be mistaken for dark matter directly; the way they can contaminate a signal is *indirectly*, through the delayed tail of that deposition—scintillator afterglow and phosphorescence, which arrives at low energy, can persist long after the muon has passed, and carries the muon's seasonal time structure into the region where a dark matter signal would appear.

My objective this term is to measure that channel directly: **how much of the low-energy background in the DM-Ice17 signal region is muon-induced, with what decay time, what fraction of it an IceCube veto removes and at what dead-time cost, and whether the surviving low-energy rate modulates annually and at what phase.**

The significance is not that DM-Ice17 will out-measure DAMA—with 17 kg it never will, against roughly two orders of magnitude more exposure at a lower threshold. Its lasting value is that it is the only place where the muon-induced contribution to a NaI signal region, and the effectiveness of an active muon veto against it, can be measured at all rather than simulated. Those two numbers are a direct input to the DM-Ice→COSINE-100 program and a design parameter for any future veto-instrumented NaI deployment. A secondary implication is archival: every published DM-Ice muon analysis stops in January 2015, two months after the photomultiplier gain was raised roughly tenfold, so most of the exposure has never been characterized at all.

2. Preparation Completed

This is a continuation, and the load-bearing infrastructure exists and is validated. I have built a stitched waveform-summary extraction over the full 2011–2022 run (2.38 billion events reduced to 136 columnar files), which solves the problem that high-gain signals span three digitizer channels with no single channel covering the full gamma-to-muon range. I built the first real DM-Ice live-time database in the group's code base—per-run active spans across 95,174 runs, giving 1635 low-gain and 2328 high-gain live-days per detector—replacing the calendar-day approximation every prior script used. I reproduced the published pulse-shape muon populations as a baseline, then diagnosed and fixed a $751\times$ over-count when that selection was transferred to the high-gain era, reaching 2.918 ± 0.038 muons/crystal/day at efficiency 0.949 and purity 0.933, statistically indistinguishable from the low-gain benchmark at 0.03σ. I hold a frozen 218-event IceCube-coincidence muon sample and a low-gain energy calibration validated by the 600 keV thallium/bismuth line. Every ingredient this term's measurement consumes—muon tags, an energy scale, live-time, and IceCube coincidence—is already in hand.

3. Detailed Research Plan

WP1 — Energy threshold and signal-region definition (weeks 1–2). One precondition has to be settled before anything else, and it is cheap. My own calibration notes flag that a feature previously fit as the 46.5 keV ^{210}Pb line is more likely the detector's trigger turn-on, its width being far too narrow for a real line at that energy. If the turn-on sits at tens of keV in this reduction, a 2–6 keV electron-equivalent (keVee) band contains too few events to work with. I will convert the low-gain energy column using the validated calibration, plot the spectrum, locate the turn-on, and define the lowest defensible analysis band from the data rather than from the literature. *Deliverable:* a measured threshold and a stated signal region, which every subsequent step inherits.

WP2 — Muon-induced background in the signal region (the headline). For each of roughly a thousand tagged muons per year per detector, I will count low-energy events in delayed windows following the muon, spanning microseconds to hours, and compare against randomized control windows drawn from the same runs. The control windows are the key methodological choice: because signal and control windows come from the same live-time, the comparison is self-normalizing, so this measurement is immune to the live-time systematic that dominates any rate or modulation fit. From the excess and its time dependence I extract the muon-induced leakage fraction into the signal region and the afterglow decay constant.

WP3 — IceCube veto rejection efficiency and dead-time cost. Using the IceCube-coincident muon sample, I will compute what fraction of the WP2 leakage a veto actually removes as a function of the veto time window, and the exposure lost to that window. This converts WP2's physics number into an operational one: how much muon-induced background an active veto eliminates, and what it costs. It also uses DM-Ice17's unique feature for the purpose it was deployed for.

WP4 — Annual modulation of the residual low-energy rate. On live-time-normalized day-of-year histograms of non-muon-tagged events inside the WP1 band, I will fit $R(t) = R_0[1 + A\cos(2\pi(t - t_0)/365)]$ per detector, with all cuts fixed before the phase is examined. Interpreted against WP2's leakage fraction, this answers whether any low-energy modulation is consistent with a muon origin, and reports an amplitude limit if the data are consistent with no modulation.

WP5 — Parallel extension and one open systematic. Running unattended alongside the above, an ATWD0-equivalent re-extraction (already written and tested on a subsample) would extend WP2 and WP4 to the 2328 high-gain live-days, where low-energy events are digitized on a channel my current reduction does not carry. This is deliberately *not* on the critical path: it is the most storage-intensive piece of the project, and the low-gain era alone supports a complete result. Separately, the 2015 high-gain muon rate sits 12.5σ below 2016–2020 and is demonstrably not a selection artifact; I will test whether per-run live-time is overestimated because it cannot resolve intra-run dead time, which must be settled before any high-gain rate claim.

Risks and fallbacks. If WP1 finds the threshold too high for a 2–6 keVee band, WP2 and WP4 move to the lowest usable band and the leakage measurement proceeds unchanged—this is why WP2 rather than the modulation fit is the headline. If the high-gain re-extraction does not finish, WP5 is simply dropped without touching the primary result. All selections live in a single canonical module so no two scripts can disagree about a cut, and IceCube coincidence remains available throughout as an independent hard label.

4. Time Frame (September–December 2026)

I plan on 10–12 hours per week; long extractions run unattended on IceCube's Cobalt and NPX systems, so compute time does not compete with coursework.

Weeks 1–2 (Sep 8–19): WP1 threshold and band definition; launch the WP5 extraction in the background. *Milestone 1: measured threshold, reviewed with my mentor.*

Weeks 3–7 (Sep 22–Oct 24): WP2 delayed-window analysis, including the control-window construction and systematic checks. *Milestone 2: muon-induced leakage fraction and decay constant.*

Weeks 8–10 (Oct 27–Nov 14): WP3 veto rejection efficiency and dead-time curve; present WP2–WP3 at group meeting.

Weeks 11–13 (Nov 17–Dec 5): WP4 modulation fit and its interpretation against the leakage fraction; fold in WP5 if the extraction has landed.

Weeks 14–15 (Dec 8–19): 2015 live-time systematic; write an internal DM-Ice/IceCube analysis note, draft a conference abstract, and submit the final report.

5. Faculty Involvement and Support Network

My faculty mentor is Prof. Carlos A. Argüelles Delgado (Department of Physics), who leads Harvard's IceCube group and has supervised this project throughout. He sets the physics scope and priorities, reviews results at each milestone, and has previously required me to freeze and defend a completed analysis in writing before moving on—a practice that produced the wrap-up memo underpinning this proposal. He supervises through weekly group meetings and regular one-on-one meetings, provides the IceCube Collaboration channel through which this work receives internal review, and provides the computing access the project depends on: IceCube's Cobalt and NPX systems and the group's GPU allocation on the FAS research cluster. An early decision I will bring to him is whether to frame the term's output primarily as a background-characterization result for the collaboration or as a modulation measurement, which affects emphasis in WP4.

Day-to-day technical supervision comes from the group's postdoctoral researcher, Felix Yu, who has defined several of the tasks I have already completed—including the simulation energy-floor characterization and the IceCube MuonFilter overlap check—and who advises on reconstruction and machine-learning methods. Beyond the group, the IceCube Collaboration's internal review process provides external scrutiny before publication, and the DM-Ice and COSINE-100 literature, especially A. J. F. Hubbard's 2015 thesis, provides the published pulse-shape baseline against which every step of my selection is validated. I also maintain written session handoffs across the three machines this analysis runs on, which keeps the work auditable by my mentor and reproducible by whoever inherits it.

Selected references. A. J. F. Hubbard, *Muon-Induced Backgrounds in the DM-Ice17 NaI(Tl) Dark Matter Detector*, PhD thesis, University of Wisconsin–Madison (2015). · J. Cherwinka *et al.* (DM-Ice), "First data from DM-Ice17," Phys. Rev. D **90**, 092005 (2014). · R. Bernabei *et al.* (DAMA/LIBRA), "First model-independent results from DAMA/LIBRA-phase2," Universe **4**, 116 (2018). · G. Adhikari *et al.* (COSINE-100), "An experiment to search for dark-matter interactions using sodium iodide detectors," Nature **564**, 83 (2018). · S. Tilav *et al.* (IceCube), "Atmospheric variations as observed by IceCube," arXiv:1001.0776 (2010).